

IMO Training – Geometry

Geometry via Complex Numbers

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Notes

Unless otherwise specified, let O be the origin, and let the smaller case letter denote the corresponding complex number on the complex plane of the vertex. For example, the complex number of point A is denoted as a .

Definition. (Purely imaginary number) $i^2 = -1$.

Theorem. (De Moivre's Theorem) $\forall x \in \mathbb{R}, n \in \mathbb{N}, (\cos x + i \sin x)^n = \cos nx + i \sin nx$.

Definition. (Polar Form, Exponential Form, Modulus, Argument) $\forall z = a + bi \in \mathbb{C}$, with $a, b \in \mathbb{R}$, $\exists r \in \mathbb{R}_{\geq 0}, \theta \in (-\pi, \pi]$ such that $z = r(\cos \theta + i \sin \theta) = e^{i\theta}$. Furthermore, $r = \sqrt{a^2 + b^2}$, $\tan \theta = \frac{b}{a}$. r is known as the modulus of z , or $|z|$, and θ is known as the argument of z , or $\arg(z)$.

Definition. (Conjugate) $\forall z = a + bi \in \mathbb{C}$, with $a, b \in \mathbb{R}$, $\bar{z} = a - bi$.

Definition. (Distance) $OZ = |z|, AB = |a - b|$. $OZ^2 = z\bar{z}, AB^2 = (a - b)(\bar{a} - \bar{b})$.

Definition. (Angle) $\angle AOB = \arg \frac{b}{a} = \arg b - \arg a$. The angle between lines AB and CD is $\arg \frac{d - c}{b - a}$.

Definition. (Straight Line) The straight line AB is denoted by $\frac{z - a}{b - a} = \frac{\bar{z} - \bar{a}}{\bar{b} - \bar{a}}$.

Definition. (Unit Circle) The unit circle is denoted by $|z| = 1$.

Theorem. (Rotation) If C is the image of rotating B with respect to A by angle θ , then $c = a + (b - a)e^{i\theta}$.

Questions

1. Find the equation of the perpendicular bisector of AB in terms of a, b .
2. If AB is a chord on the unit circle, find the equation of AB in terms of a, b .
3. If A is a point on the unit circle, find the equation of the tangent line at A to the unit circle. Furthermore, if B is another point on the unit circle, find the corresponding complex number of the intersecting point of the tangents at A and B to the unit circle.
4. Consider $\triangle ABC$ with A, B, C on the unit circle. Find the corresponding complex number of the orthocentre H of $\triangle ABC$.
5. What is the necessary and sufficient condition on a, b, c, d for AB and CD to be perpendicular?
6. What is the necessary and sufficient condition on a, b, c for $\triangle ABC$ to be equilateral?
7. What is the necessary and sufficient condition on a, b, c, d for quadrilateral $ABCD$ to be cyclic?

Examples

1. $ABCD$ and $EFGH$ are straight lines, with $ABFE$, $BCGF$ and $CDHG$ being three identical squares. Find $\angle AFE + \angle AGF + \angle AHG$.
2. Squares $ABPQ$ and $ACRS$ are drawn outwardly on the sides of $\triangle ABC$. Prove that QS is twice as long as the median AM to BC .
3. Let M be the centroid of $\triangle ABC$. Suppose MB is rotated about M through an angle of 120° to take B to the point P and that MC is rotated about M through 240° to take C to Q . Prove that $\triangle APQ$ is equilateral.
4. On the sides of a triangle ABC draw externally the squares $ABMM'$, $ACNN'$ and $BCPP'$. Let A' , B' and C' be the midpoints of the segments $M'N'$, $P'M$ and PN respectively. Prove that $\triangle ABC$ and $\triangle A'B'C'$ have the same centroid.
5. Let ABC be an acute triangle. On the same side of line AC as point B draw isosceles triangles DAB , BCE and AFC with right angles at A , C and F respectively. Prove that D , E , F are collinear.
6. Prove that if equilateral triangles are erected outwardly on the sides of a triangle, then their centres are vertices of an equilateral triangle.
7. Let A, B, C, D be four point in a plane. Prove that $AB \times CD + BC \times AD \geq AC \times BD$, with equality holds iff $ABCD$ is cyclic.
8. (CHKMO 2010) Let ABC be an arbitrary triangle. A regular n -gon is constructed outward on the three sides of $\triangle ABC$. Find all n such that the triangle formed by the three centres of the n -gons is equilateral.
9. (IMO 2012 P1) Given triangle ABC the point J is the centre of the excircle opposite the vertex A . This excircle is tangent to the side BC at M , and to the lines AB and AC at K and L , respectively. The lines LM and BJ meet at F , and the lines KM and CJ meet at G . Let S be the point of intersection of the lines AF and BC , and let T be the point of intersection of the lines AG and BC . Prove that M is the midpoint of ST .
10. (IMO 2023 P2) Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC . Let S be the midpoint of the arc CB of Ω containing A . The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L . Denote the circumcircle of triangle BDL by ω . Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.